

Laptops Segmentation into seperate clusters using KMeans

In [183]:

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import (
    LabelEncoder,
    StandardScaler,
    OneHotEncoder
)

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

#Ignore all incoming warnings (if any):
from warnings import filterwarnings as f
f("ignore")

df = pd.read_csv("laptopPrice.csv")
```

In [184]:

```
print(df.head())
```

	brand	processor_brand	processor_name	processor_gnrtn	ram_gb	ram_type	\
0	ASUS	Intel	Core i3	10th	4 GB	DDR4	
1	Lenovo	Intel	Core i3	10th	4 GB	DDR4	
2	Lenovo	Intel	Core i3	10th	4 GB	DDR4	
3	ASUS	Intel	Core i5	10th	8 GB	DDR4	
4	ASUS	Intel	Celeron Dual	Not Available	4 GB	DDR4	

	ssd	hdd	os	os_bit	graphic_card_gb	weight	warranty	\
0	0 GB	1024 GB	Windows	64-bit	0 GB	Casual	No warranty	
1	0 GB	1024 GB	Windows	64-bit	0 GB	Casual	No warranty	
2	0 GB	1024 GB	Windows	64-bit	0 GB	Casual	No warranty	
3	512 GB	0 GB	Windows	32-bit	2 GB	Casual	No warranty	
4	0 GB	512 GB	Windows	64-bit	0 GB	Casual	No warranty	

	Touchscreen	msoffice	Price	rating	Number of Ratings	Number of Reviews
0	No	No	34649	2 stars	3	0
1	No	No	38999	3 stars	65	5
2	No	No	39999	3 stars	8	1
3	No	No	69990	3 stars	0	0
4	No	No	26990	3 stars	0	0

In [185]:

```
print(df.shape)
```

(823, 19)

In [186]:

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 823 entries, 0 to 822
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   brand                 823 non-null    object
```

```
1 processor_brand      823 non-null object
2 processor_name       823 non-null object
3 processor_gnrtn      823 non-null object
4 ram_gb               823 non-null object
5 ram_type             823 non-null object
6 ssd                  823 non-null object
7 hdd                  823 non-null object
8 os                   823 non-null object
9 os_bit               823 non-null object
10 graphic_card_gb     823 non-null object
11 weight              823 non-null object
12 warranty            823 non-null object
13 Touchscreen         823 non-null object
14 msoffice             823 non-null object
15 Price               823 non-null int64
16 rating              823 non-null object
17 Number of Ratings   823 non-null int64
18 Number of Reviews   823 non-null int64
```

```
dtypes: int64(3), object(16)
```

```
memory usage: 122.3+ KB
```

```
None
```

```
In [187]:
```

```
df.isnull().sum()
```

```
Out[187]:
```

```
brand                0
processor_brand      0
processor_name       0
processor_gnrtn      0
ram_gb               0
ram_type             0
ssd                  0
hdd                  0
os                   0
os_bit               0
graphic_card_gb     0
weight              0
warranty            0
Touchscreen         0
msoffice             0
Price               0
rating              0
Number of Ratings   0
Number of Reviews   0
dtype: int64
```

```
In [188]:
```

```
# percentage missing
```

```
(df.isnull().sum()/len(df))*100
```

```
Out[188]:
```

```
brand                0.0
processor_brand      0.0
processor_name       0.0
processor_gnrtn      0.0
ram_gb               0.0
ram_type             0.0
ssd                  0.0
hdd                  0.0
os                   0.0
os_bit               0.0
graphic_card_gb     0.0
weight              0.0
warranty            0.0
Touchscreen         0.0
msoffice             0.0
Price               0.0
```

```
rating          0.0
Number of Ratings 0.0
Number of Reviews 0.0
dtype: float64
```

In [189]:

```
df.duplicated().sum()
```

Out[189]:

```
np.int64(21)
```

In [190]:

```
# remove duplicates
df.drop_duplicates(inplace=True)
```

In [191]:

```
df.columns = df.columns.str.lower()
```

```
df.head()
```

```
# RAM conversion
```

```
df["ram_gb"] = (
    df["ram_gb"]
    .str.replace(" GB", "")
    .astype(int)
)
```

```
# SSD conversion
```

```
df["ssd"] = (
    df["ssd"]
    .str.replace(" GB", "")
    .astype(int)
)
```

```
# HDD conversion
```

```
df["hdd"] = (
    df["hdd"]
    .str.replace(" GB", "")
    .astype(int)
)
```

```
# Graphic card
```

```
df["graphic_card_gb"] = (
    df["graphic_card_gb"]
    .str.extract("(\\d+) ")
    .fillna(0)
    .astype(int)
)
```

```
# Rating conversion
```

```
df["rating"] = (
    df["rating"]
```

```

        .str.replace(" stars", "", regex=False)
        .str.replace(" star", "", regex=False)
        .astype(int)
    )

# OS bit

df["os_bit"] = (
    df["os_bit"]
    .str.replace("-bit", "")
    .astype(int)
)

df.head()

# =====
# 7. CREATE NEW FEATURES
# =====

# Total storage

df["total_storage"] = (
    df["ssd"] + df["hdd"]
)

# Storage type

df["has_ssd"] = np.where(
    df["ssd"] > 0, 1, 0
)

# GPU availability

df["has_gpu"] = np.where(
    df["graphic_card_gb"] > 0, 1, 0
)

# Laptop performance score

df["performance_score"] = (
    df["ram_gb"] * 2 +
    df["processor_name"].map({
        "Core i3": 3,
        "Core i5": 5,
        "Core i7": 7,
        "Core i9": 9,
        "Ryzen 3": 3,
        "Ryzen 5": 5,
        "Ryzen 7": 7,
        "M1": 8,
        "Celeron Dual": 2,
        "Pentium Quad": 2
    }).fillna(0)
    +
    df["graphic_card_gb"]
)

# customer popularity

```

```

df["popularity_score"] = (
    df["number of ratings"] +
    df["number of reviews"]
)

# Total Storage:
df["total_storage"] = (
    df["ssd"] + df["hdd"]
)

#GPU availability:
df["has_gpu"] = np.where(
    df["graphic_card_gb"]>0,
    1,
    0
)

df.drop(
    [
        "processor_name"
    ],
    axis=1,
    inplace=True
)

```

In []:

Problem statement : "A laptop company wants to segment its laptops into different customer segments based on specifications, pricing, and customer feedback. Build a clustering model that groups laptops into meaningful categories such as Budget laptops, Professional laptops, Gaming laptops, and Premium laptops."

In [192]:

```
df.shape
```

Out[192]:

```
(802, 23)
```

In [193]:

```

cluster_df = df[
    [
        "price",
        "ram_gb",
        "graphic_card_gb",
        "total_storage",
        "performance_score",
        "rating",
        "number of ratings",
        "number of reviews",
        "brand",
        "processor_brand",
        "os"
    ]
]

cat_cols = cluster_df.select_dtypes(include="object").columns
num_cols = cluster_df.select_dtypes(exclude="object").columns

preprocessor = ColumnTransformer(
    transformers=[

```

```

    (
        "categorical",
        OneHotEncoder(handle_unknown="ignore"),
        cat_cols
    ),
    (
        "numeric",
        StandardScaler(),
        num_cols
    )
]
)

X = preprocessor.fit_transform(cluster_df)

```

In [194]:

```

from sklearn.cluster import KMeans

wcss = []

for k in range(1,11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    kmeans.fit(X)

    wcss.append(kmeans.inertia_)

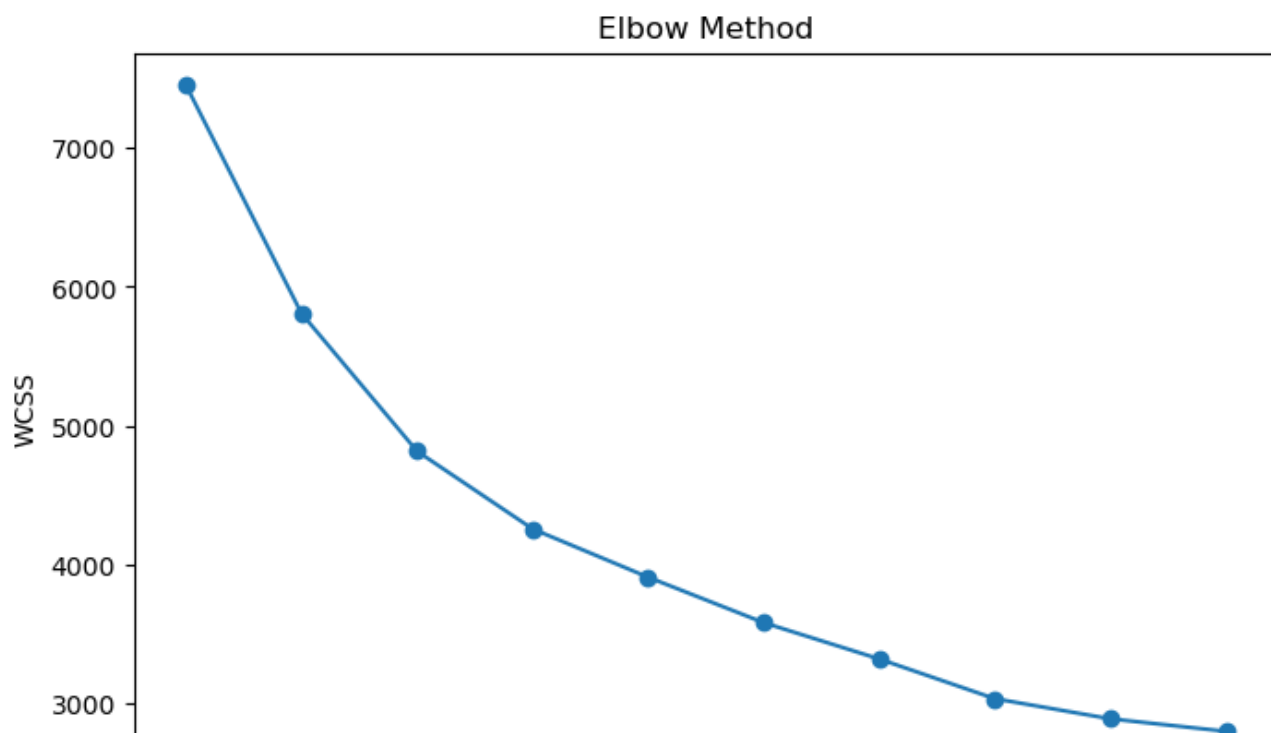
plt.figure(figsize=(8,5))

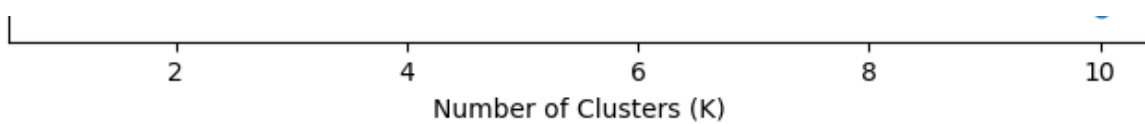
plt.plot(
    range(1,11),
    wcss,
    marker="o"
)

plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")

plt.show()

```





In [195]:

```
from sklearn.metrics import silhouette_score

for k in range(2,11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    labels = kmeans.fit_predict(X)

    score = silhouette_score(X, labels)

    print(f"K = {k} --> Silhouette Score = {score:.4f}")
```

```
K = 2 --> Silhouette Score = 0.3039
K = 3 --> Silhouette Score = 0.3262
K = 4 --> Silhouette Score = 0.1927
K = 5 --> Silhouette Score = 0.2006
K = 6 --> Silhouette Score = 0.2071
K = 7 --> Silhouette Score = 0.2041
K = 8 --> Silhouette Score = 0.2084
K = 9 --> Silhouette Score = 0.2163
K = 10 --> Silhouette Score = 0.2091
```

In [196]:

```
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

clusters = kmeans.fit_predict(X)

cluster_df["cluster"] = clusters
```

In [197]:

```
pd.crosstab(
    cluster_df["cluster"],
    cluster_df["brand"]
)
```

Out[197]:

brand	APPLE	ASUS	Avita	DELL	HP	Lenovo	MSI	acer
cluster								
0	9	175	5	134	95	117	27	40
1	12	58	1	15	35	26	23	10
2	2	10	0	0	6	1	0	1

In [198]:

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X)
```

```
plt.figure(figsize=(10,6))

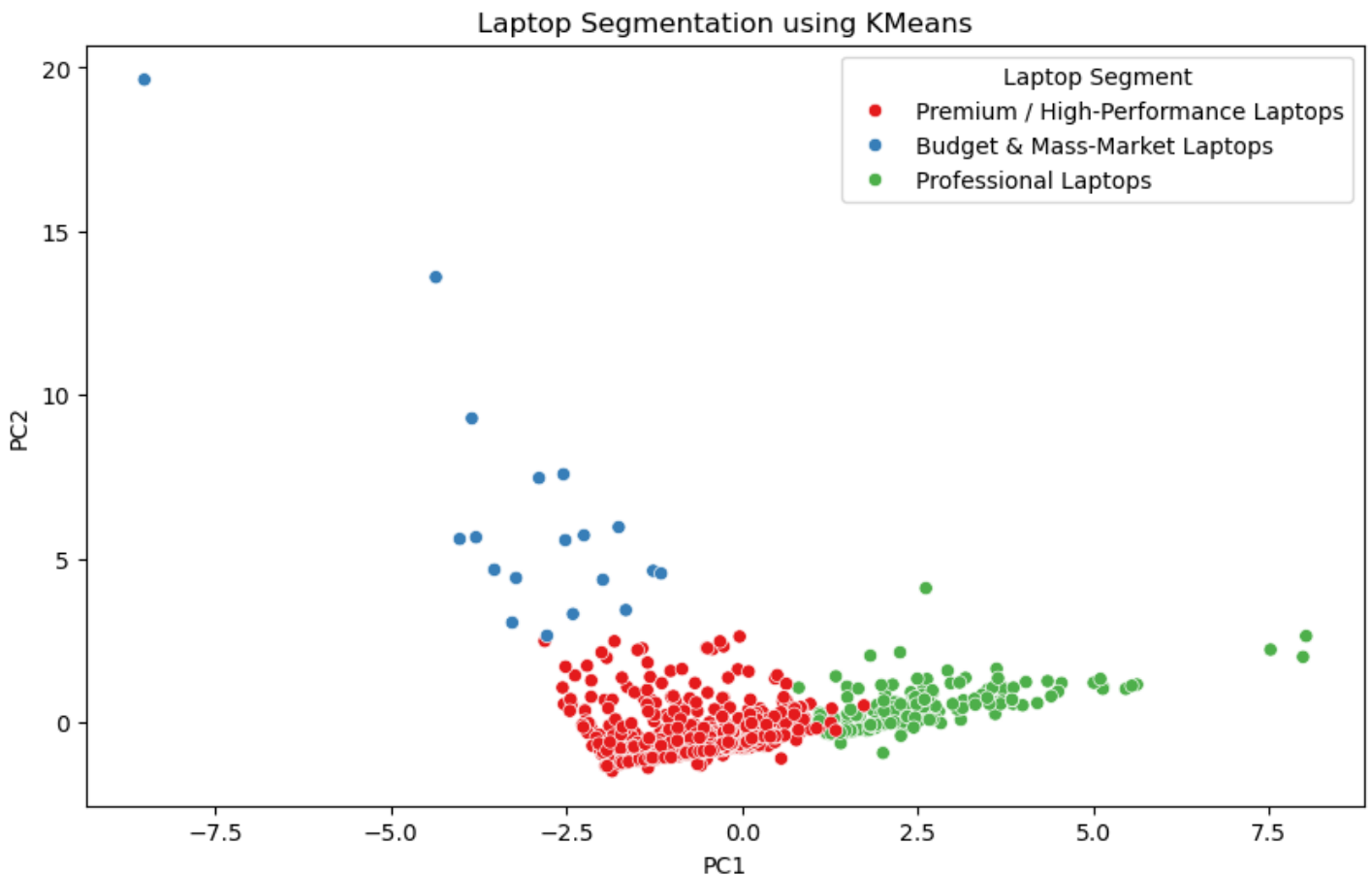
cluster_names = {
    0: "Premium / High-Performance Laptops",
    1: "Professional Laptops",
    2: "Budget & Mass-Market Laptops"
}

cluster_labels = pd.Series(clusters).map(cluster_names)

sns.scatterplot(
    x=X_pca[:,0],
    y=X_pca[:,1],
    hue=cluster_labels,
    palette="Set1"
)

plt.xlabel("PC1")
plt.ylabel("PC2")

plt.legend(title="Laptop Segment")
plt.title("Laptop Segmentation using KMeans")
plt.show()
```



Although the business initially expected four segments (Budget, Professional, Gaming, Premium), Silhouette Analysis indicated that the optimal number of clusters is 3 (score = 0.3278). Therefore, K-Means with K=3 was selected to obtain the most meaningful and well-separated customer segments.

Now we will explore what type of information do each cluster holds and draw some insights based on it:

In [199]:


```
cluster_profile = cluster_df.groupby("cluster").agg({
    "price": "mean",
    "ram_gb": "mean",
    "graphic_card_gb": "mean",
    "total_storage": "mean",
    "performance_score": "mean",
    "rating": "mean",
    "number of ratings": "mean",
    "number of reviews": "mean"
}).round(2)

cluster_profile
```

Out[199]:

	price	ram_gb	graphic_card_gb	total_storage	performance_score	rating	number of ratings	number of reviews
cluster								
0	63649.39	6.57	0.86	651.91	18.28	3.58	198.92	24.28
1	123263.53	15.96	2.43	795.73	40.77	3.51	81.94	11.42
2	47465.95	6.40	0.90	678.40	17.65	3.80	5298.70	613.50

In [200]:

```
pd.crosstab(
    cluster_df["cluster"],
    cluster_df["brand"]
)
```

Out[200]:

brand	APPLE	ASUS	Avita	DELL	HP	Lenovo	MSI	acer
cluster								
0	9	175	5	134	95	117	27	40
1	12	58	1	15	35	26	23	10
2	2	10	0	0	6	1	0	1

In [201]:

```
pd.crosstab(
    cluster_df["cluster"],
    cluster_df["os"]
)
```

Out[201]:

os	DOS	Mac	Windows
cluster			
0	10	9	583
1	5	12	163
2	1	2	17

Executive Summary

Using K-Means clustering (K=3), laptops were segmented into three distinct customer-oriented groups based on:

- Specifications
- Price
- Customer feedback

The three segments identified are:

Cluster	Segment Name
0	Professional Laptops
1	Premium / High-Performance Laptops
2	Budget & Mass-Market Laptops

Cluster 0: Professional Laptops

Characteristics

Metric	Value
Avg Price	₹63,649
Avg RAM	6.57 GB
Avg GPU	0.86 GB
Avg Storage	652 GB
Avg Performance Score	18.28
Avg Rating	3.58

Brand Distribution

Dominated by:

- ASUS (175)
- Dell (134)
- Lenovo (117)
- HP (95)

Interpretation

These laptops provide:

- Moderate pricing
- Standard RAM
- Basic graphics capability
- Adequate storage

Target audience:

- Office professionals
- College students
- Small businesses

Business Insight

This is the largest mainstream segment and likely contributes a significant share of sales volume.

Cluster 1: Premium / High-Performance Laptops

Characteristics

Metric	Value
Avg Price	₹123,264
Avg RAM	15.96 GB

Metric	Value
Avg Price	₹47,466
Avg Storage	796 GB
Avg Performance Score	40.77
Avg Rating	3.51

Brand Distribution

Strong presence of:

- ASUS
- Apple
- MSI

Interpretation

These laptops have:

- Highest prices
- Highest RAM
- Highest GPU capability
- Best performance scores

Target audience:

- Gamers
- Developers
- Video editors
- Power users

Business Insight

Although smaller in size, this segment generates the highest revenue per laptop.

Cluster 2: Budget & Mass-Market Laptops

Characteristics

Metric	Value
Avg Price	₹47,466
Avg RAM	6.40 GB
Avg GPU	0.90 GB
Avg Storage	678 GB
Avg Performance Score	17.65
Avg Rating	3.80
Avg Ratings Count	5298
Avg Reviews Count	614

Interpretation

These laptops are:

- Lowest priced
- Basic specifications
- Highest customer engagement
- Most reviewed products

Business Insight

This segment appears to be the mass-market segment with the highest demand and visibility.

The exceptionally high number of ratings and reviews indicates strong market adoption.

Customer Feedback Analysis

Most Popular Segment

Cluster 2

Average:

- 5299 ratings
- 614 reviews

This suggests:

- High sales volume
- Strong online visibility
- Large customer base

Highest Rated Segment

Cluster 2

Average Rating:



Although specifications are modest, customers appear satisfied because expectations align with pricing.

Operating System Insights

Windows Dominates

Across all clusters:

- Majority of laptops use Windows.

This indicates:

- Windows remains the primary platform across all customer segments.

Mac Devices

Small presence in Premium Cluster.

This aligns with Apple's positioning as a premium brand.

Strategic Recommendations

Budget Segment

Focus on:

- Student offers
- Online marketplaces

- Volume-based sales

Professional Segment

Focus on:

- Corporate customers
- Productivity features
- Extended warranties

Premium Segment

Focus on:

- Gaming
- Creative professionals
- Premium branding

Final Conclusion

K-Means successfully identified three natural laptop segments:

- 1. **Budget & Mass-Market Laptops** → High demand, affordable devices.
- 2. **Professional Laptops** → Balanced performance and pricing.
- 3. **Premium / High-Performance Laptops** → Expensive, powerful devices targeting advanced users.

The silhouette analysis showed that **K = 3 (0.3278)** produced the most meaningful segmentation, indicating that the laptop market in this dataset naturally forms three major customer-oriented groups rather than four.

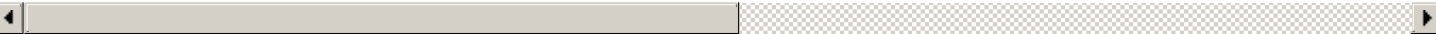
In [202]:

```
df.head(2)
```

Out[202]:

	brand	processor_brand	processor_gnrtn	ram_gb	ram_type	ssd	hdd	os	os_bit	graphic_card_gb	...	msoffice
0	ASUS	Intel	10th	4	DDR4	0	1024	Windows	64	0	...	No
1	Lenovo	Intel	10th	4	DDR4	0	1024	Windows	64	0	...	No

2 rows x 23 columns



In []: